Skilled Immigration and Business Diversity: Evidence from the H-1B Lottery*

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Abstract

I examine the effect of H-1B visa lottery outcomes on firm concentration and business diversity outcomes at the county level in the United States, leveraging a natural experiment. I combine H-1B data from the Department of Labor and US Citizenship and Immigration Services with data from the US Census Bureau's County Business Patterns to analyze changes in firm concentration and business diversity. I measure business diversity using two indices: the Herfindahl-Hirschman Index and the Shannon Diversity Index. Using a continuous difference-in-differences design, I find that counties with a higher lottery win rate experienced a 0.64% decline in the Herfindahl-Hirschman Index and a 0.27% increase in the Shannon Diversity Index. These findings suggest that high-skilled immigration contributes to economic diversification, supporting the broader argument that the H-1B visa plays a role beyond filling labor shortages.

JEL Classification: J61, J68, L11

Keywords: H-1B visa, Skilled Immigration, Firm Concentration, Business Diversity, Entrepreneurship, New Business Formation

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1 Introduction

High-skilled immigration has long played a critical role in shaping the US economy, particularly in sectors that rely on innovation and technical expertise, such as the tech industry and semiconductor manufacturing. The H-1B program, which admits tens of thousands of high-skilled foreign workers every year, is central to this process. While much of the existing research on the H-1B visa is focused on wages and employment of domestic workers, firm productivity outcomes, and city-level employment and wage outcomes, relatively little is known about how the H-1B visa affects the structure of local economies. In particular, the relationship between skilled foreign labor and business diversity - the variety and concentration of firms in a given area - remains understudied.

This paper investigates whether exposure to skilled immigration affects business diversity and firm concentration at the county level. A more diverse business environment enhances economic resilience, generates welfare insurance gains (de Soyres et al., 2024), and fosters entrepreneurship. Business diversity is also a strong driver of employment growth (Frenken, Van Oort and and Verburg, 2007) and economic growth (Glaeser, Kerr and Ponzetto, 2010). This suggests that business diversity plays a critical role for long-term economic development and resilience. Understanding how skilled immigration affects local firm dynamics has implications for both immigration policy and local economic policy.

To address this question, I exploit the randomized allocation of visas in the 2016 H-1B lottery using a continuous difference in differences design. I begin by constructing a proxy of H-1B lottery success at the county level by matching data from the US Citizenship and Immigration Services' H-1B Employer Datahub (U.S. Citizenship and Immigration Services, 2016) with data from Labor Condition Applications (LCA) (U.S. Department of Labor, Office of Foreign Labor Certification, 2016) for the 2016 H-1B lottery. I take several steps to make sure that the number of intended applications matches the number of true applications. I use this data to construct a proxy of the H-1B lottery win rate by dividing the number of accepted applications from the USCIS dataset by the total number of applications from the LCA dataset. Next, I merge my win rate measure to a merged panel dataset of the US Census Bureau County Business Patterns (U.S. Census Bureau, 2025) with 4-digit NAICS codes from the years of 2012-2022. I then construct two measures of business diversity from the County Business Patterns dataset, a Herfindahl-Hirschman Index and a Shannon Diversity Index.

I conduct an event study analysis to evaluate whether the parallel trends assumption is satisfied. Importantly, I found that counties with higher and lower win rates in the 2016 H-1B lottery were not trending differently before the 2016 H-1B lottery. Moving on to the main results, I find that a ten percentage point increase in the H-1B lottery win rate is associated with a 0.64 percent decline in the Herfindahl-Hirschman Index and a 0.27 percent increase in the Shannon Diversity Index in a given county. These effects are robust to several alternative specifications, including placebo tests, exclusion of post-COVID years, and alternative standard error treatments. A mediation analysis suggests that most of the gains in business diversity are driven by increased business formation following the H-1B lottery. Heterogeneity analysis reveals that these effects are especially strong in immigrant-intensive sectors, with high-skilled immigrant-intensive industries showing the largest gains. Together, these findings suggest that skilled immigration not only contributes to firm-level outcomes but also to the broader structure of local economies.

2 Overview of the H-1B Visa

The H-1B visa, created under the Immigration Act of 1990, is the primary pathway through which US firms hire foreign skilled labor. The program is capped annually at 85,000 new visas, with 65,000 visas offered under the regular cap and 20,000 visas for the advanced degree exemption (for applicants with a master's degree or higher from US institutions). When the number of applications exceeds the cap, the US government conducts a randomized lottery to allocate visas among eligible applications. Non-profit firms, such as universities, research institutions, and those affiliated with universities, are generally exempt from the cap. In contrast, private-sector for-profit firms aiming to hire high-skilled foreign labor are subject to the cap and must enter the lottery.

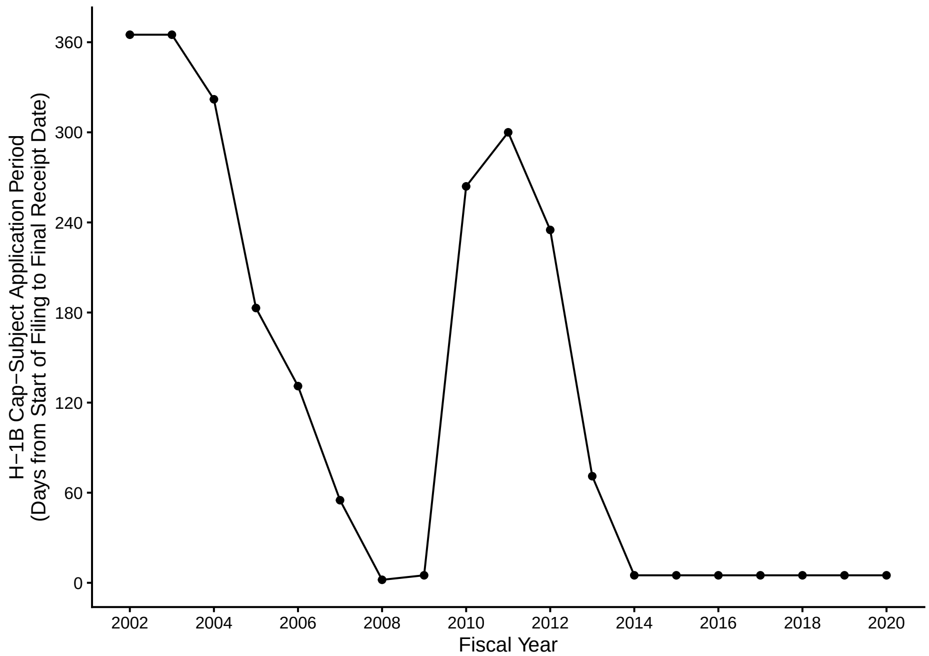

Figure 1: Number of days until the H-1B cap was exceeded. Data taken from press releases of the US Citizenship and Immigration Services (U.S. Citizenship and Immigration Services, Various years).

From 2009 to 2012, there was not enough demand for the H-1B visa for all the visas to be subject to the lottery, but for the FY2014 lottery, which took place in 2013, all H-1B visas were subject to the lottery. Prior to 2013, the H-1B visas were distributed on a first-come-first-serve basis, but after 2013 and in years where the H-1B cap was exceeded early, such as in the FY2008 lottery examined by Mahajan et al. (2024), the application period was open for 5 days and all petitions deemed valid were subject to the H-1B lottery.

This lottery mechanism introduces quasi-random variation in visa allocation, which has been leveraged in literature as a natural experiment to estimate the causal effects of high-skilled immigration. Because the lottery determines which firms are permitted to employ H-1B workers in a given fiscal year, the win rate of a firm and the subsequent county it is located in will determine how much a county is exposed to high-skilled foreign labor. This provides an opportunity to estimate the impact of skilled immigration on local economic outcomes.

3 Literature Review

Studies on the H-1B lottery, such as Mahajan et al. (2024) and Doran, Gelber and Isen (2022) use exogenous variation in the H-1B lottery win rate as an independent variable. Oftentimes, their choice of identification is a difference-in-differences (DiD) strategy because the quasi-random nature of the H-1B lottery creates plausible exogenous variation. Although Clemens and Lewis (2022) study low-skilled immigration through the H-2B visa, the H-2B visa also requires a visa lottery. Clemens and Lewis (2022) uses DiD as well as a two-stage least squares regression as their empirical strategy. The lottery papers all use similar data sources, usually using USCIS H-1B visa approvals combined with the Department of Labor's records of all H-1B visa applications, or using H-1B approvals requested through FOIA (Mahajan et al., 2024; Clemens and Lewis, 2022; Doran, Gelber and Isen, 2022).

Although the H-1B lottery is random, selection into the lottery may not be. There may be endogeneity in the firms that enter the lottery, which may bias estimates, especially in the denominator, for H-1B lottery papers when they construct their win-rate measure. Another issue that affects all H-1B lottery papers is violations of the stable unit treatment value assumption (SUTVA) as the H-1B visa lottery is an annual event. There is a strong possibility that firms being studied were treated by the previous year's H-1B lottery, which may cause attenuation bias. This means that the true estimates are likely larger than the measured estimates.

While H-1B lottery papers often focus on labor market outcomes such as wages, employment, or firm productivity, a growing literature assesses the role of immigration on entrepreneurship and new business formation. Immigrants are more likely than natives to start businesses (Pekkala Kerr and Kerr, 2020) and through business formation, contribute more to labor demand (Azoulay et al., 2022; Peri, Shih and Sparber, 2015). High-skilled immigrant exposure in American masters programs contributes to greater startup formation (Beine, Peri and Raux, 2024). However, this literature has largely overlooked the effect of immigration on broader county-level measures of business diversity or firm concentration.

To quantify the structure and variety of businesses, researchers often rely on metrics such as the Herfindahl-Hirschman Index, which is frequently used by the Department of Justice and the Federal Reserve to capture the competitive effects of mergers. HHI is calculated by squaring the market share of all firms in a market and then summing up the squares, typically in a defined market (Rhoades, 1993). However, the HHI can be repurposed to reflect firm diversity across sectors at the regional level (Autor et al., 2020; de Soyres et al., 2024). Beyond diversity, the HHI has also been used to measure the effect of monopsony on the wages of guest workers on the H-1B, H-2A, and H-2B visas (Gibbons et al., 2019; Kim and Pei, 2022), reinforcing its flexibility as a structural economic indicator. A lower HHI typically signals a more diverse set of economic activities and is known to make regions more resilient to economic shocks (de Soyres et al., 2024) and drive economic growth (Glaeser, Kerr and Ponzetto, 2010).

Recent work connecting the H-1B lottery to local economic diversity is limited. While studies such as Mahajan et al. (2024) and Mayda et al. (2023) focus on employment, wages, and firm productivity, my study builds on their identification strategies and applies them to a novel outcome. Specifically, I use an industry-level Herfindahl-Hirschman Index, calculated at the county level using 4-digit NAICS codes, to measure how exogenous shifts in the skilled labor supply from the H-1B visa lottery affect local business diversity. This approach captures structural changes in regional firm dynamics and contributes new evidence to the literature on skilled immigration and local economic dynamism.

4 Underlying Mechanisms

One central mechanism linking skilled immigration to greater local business diversity is new business formation. Prior research documents a strong link between high-skilled immigration and increased firm formation, particularly through immigrant entrepreneurship and spillover effects. In general, immigrants enter entrepreneurship at higher rates than natives and immigrant-owned firms are less likely to exit compared to non-immigrant firms (Pekkala Kerr and Kerr, 2020; Fairlie and Lofstrom, 2015). In addition, high-skilled immigration contributes to the creation of innovative start-ups — an increase of 100 foreign Master's graduates in a university cohort leads to an increase of 1 additional startup (Beine, Peri and Raux, 2024). My results suggest that counties with higher H-1B visa lottery success rates experience an increase in the number of businesses following the H-1B lottery. This, in turn, mediates much of the observed rise in business diversity and decline in firm concentration. Consistent with this mechanism, my mediation analysis suggests that increased business formation accounts for a substantial share of the observed rise in business diversity following the H-1B lottery.

New firm entry may arise through several channels. First, high-skilled immigrants may contribute to business formation, either by founding firms themselves or creating spillover effects that allow native-born founders to start firms. H-1B visa holders tend to raise the productivity and growth potential of firms (Doran, Gelber and Isen, 2022; Mahajan et al., 2024), which may enable dynamic firm ecosystems. Second, spousal entrepreneurship and community spillovers may provide a supporting role. H-1B visa holders are often accompanied by their spouses on H-4 visas, many of whom are also highly educated and have prior work experience. As H-4 visa holders may face employment restrictions, especially without employment authorization documents (EADs), they may still pursue entrepreneurship as an alternative economic activity. This contributes to firm entry primarily through service oriented or low-barrier industries.

Third, agglomeration effects may further amplify business diversity. Higher concentrations of high-skilled immigrants can foster localized knowledge spillovers, improve labor market matching, and increase local income, all of which may generate demand for a wider range of goods and services. Finally, increases in consumer income and market size due to H-1B inflows may also contribute to greater firm variety. H-1B visa holders, who are subject to minimum salary requirements, are likely to increase aggregate local income, further boosting firm entry.

Taken together, these mechanisms are consistent with the finding that skilled immigration modestly increases business diversity at the county level, with new firm entry playing a substantial mediating role.

5 Data

I construct a balanced panel of counties from 2012-2022 for which complete NAICS-level establishment counts are available. I then merge this with a county-level win-rate measure derived from the 2016 H-1B visa lottery to create the final dataset used in analysis.

5.1 Business Diversity Data - US Census Bureau

My business diversity data comes from the US Census Bureau's County Business Patterns dataset (U.S. Census Bureau, 2025) from the years 2012-2022 using 4-digit NAICS codes. NAICS codes (North American Industry Classification System) are used to classify businesses by industry. 4-digit NAICS codes are used because they provide enough granularity for my analysis without unnecessarily increasing dataset size. Data from 2012-2022 is used based on data availability. Years prior to 2012 are not stored on data.census.gov and are subject to removal by executive order. To clean and merge all the cross-sectional data, I follow these steps. First, I upload the datasets separately because County Business Patterns datasets from 2012-2013, 2014-2016, and 2017-2022 all have different structures. I exclude counts of small businesses (under 500 employees) to maintain consistency across years, since this variable was not reported in the 2012-2013 data and it double-counts the total number of businesses in each County Business Patterns cross-sectional dataset. I also used a 2012-2017 NAICS code crosswalk to convert the 2012 NAICS codes from the 2012-2016 datasets to 2017 NAICS codes.

To evaluate the level of business diversity and firm concentration at the county level, I construct two measures using data from the US Census Bureau: an industry-level Herfindahl-Hirschman Index and an industry-level Shannon Diversity Index. The Herfindahl-Hirschman Index, which measures industry concentration, is given as follows:

$$\lambda = 10,000 \sum_{i=1}^{R} p_i^2$$

where each $p_i = \frac{\text{Establishments}_i}{\text{Total Establishments in County-Year}}$ is the share of establishments in NAICS category i and R is the number of distinct NAICS categories observed in a specific county-year. The industry-level HHI can be decreased if new firm entry occurs across a broader range of NAICS categories, diluting the dominance of a single industry. A higher HHI indicates greater firm concentration (i.e. less diversity), while a lower HHI indicates a more diversified industry structure. For example, Fairfax County, Virginia ranks near the median (50th percentile) in terms of diversity across all US counties with an HHI of 316, Santa Clara County, California, the home of Silicon Valley, ranks around the 78th percentile with an HHI of 219, while Nantucket County, Massachusetts is close to the 25th percentile with an HHI of 563. Figure 2 shows every US county by HHI, measured in percentiles for my sample.

In addition to serving as a descriptive measure of firm concentration at the county level, I use the HHI as a dependent variable in my analysis to examine how high-skilled immigration may influence the structure of local economies. By regressing HHI on H-1B lottery win rates, I test whether counties with greater exposure to high-skilled immigrant labor exhibit shifts towards diversified industry structures. This approach differs from de Soyres et al. (2024), who use HHI as an independent variable to explain labor market resilience; in my setting, HHI serves as an endogenous outcome potentially influenced by changes in firm concentration

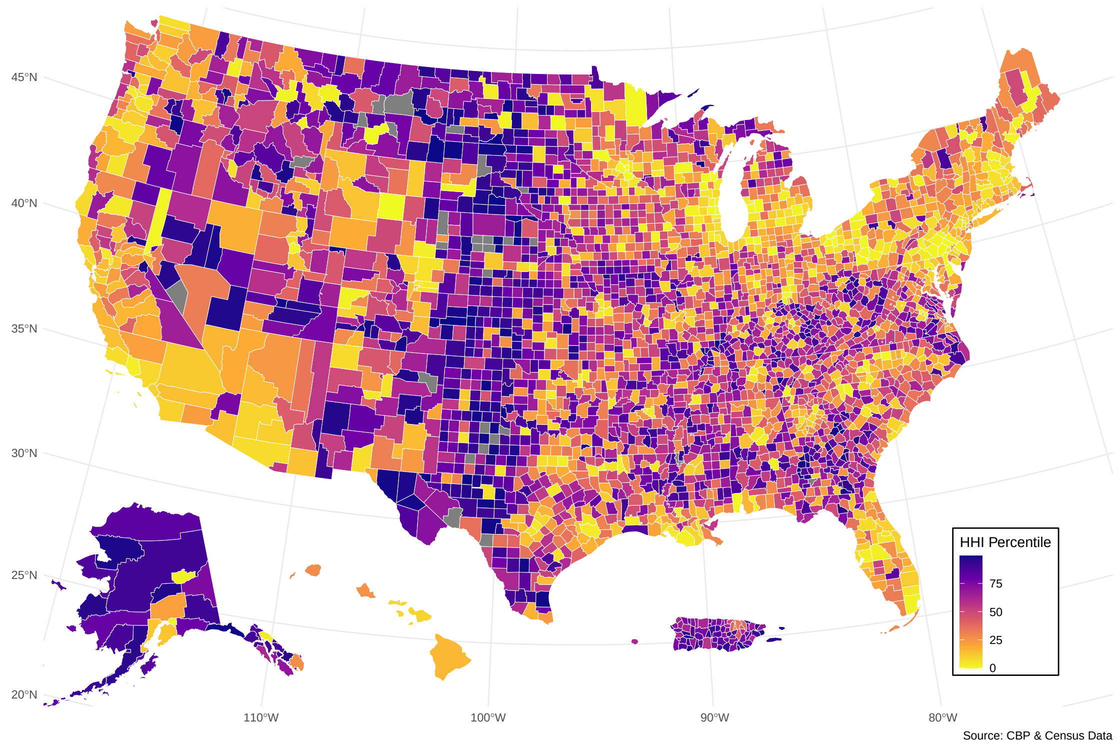

Figure 2: All US Counties by HHI Percentile in 2022. Data from County Business Patterns (U.S. Census Bureau, 2025) and author's calculations.

following immigration shocks.

The industry-level Shannon Diversity Index, which measures industry diversity, is given as follows:

$$H' = -\sum_{i=1}^{R} p_i \log(p_i)$$

Unlike the HHI, the Shannon Index increases with diversity: higher values indicate a more even distribution of establishments across NAICS sectors. Together, the HHI and Shannon Index offer complementary views of industry structure and business diversity. While the HHI captures concentration, the Shannon Index captures richness (the number of businesses in each NAICS category) and evenness (how evenly businesses are distributed across NAICS categories).

5.2 H-1B Data - USCIS H-1B Approvals and LCAs

My H-1B data comes from two different sources. First, I obtain firm-level data on H-1B labor that won the 2016 H-1B lottery from the US Citizenship and Immigration Services' H-1B Employer Data Hub (U.S. Citizenship and Immigration Services, 2016). I choose 2016 because of data limitations on accessing County Business Patterns before 2012. Choosing 2016 as my treatment year allows for a balanced panel with five years of data before and after the treatment year, enabling cleaner event study estimation and a more symmetric analysis of pre- and post-trends. The 2016 H-1B lottery was also very typical of H-1B lotteries with a similar application cycle and overall win rate as other lotteries after 2013, as shown in Appendix Figure A1. The dataset includes annual records of the counts of employees who won the H-1B lottery at the firm level, containing information on the employer name, worksite location by city, state, and zip code, and a count of the number of workers employed. Since the dataset reports zip code locations, I use a zip-to-county crosswalk from the Department of Housing and Urban Development (U.S. Department of Housing and Urban Development, 2016) to create a "county" column for merging purposes later. After assigning counties to each zip code, I aggregate the total number of H-1B lottery winners at the county level. I remove columns not used in analysis, such as continuing approvals and denials, initial denials, employer names, tax IDs, and industry codes.

I also use records of Labor Condition Applications (LCA) for the 2016 H-1B lottery from the Department of Labor's Office of Foreign Labor Certification (U.S. Department of Labor, Office of Foreign Labor Certification, 2016). This dataset includes records on every application for the H-1B visa for the 2016 H-1B lottery, containing information on the date of filing, employment start and end dates, employer names and addresses, the employer postal code, and the job title. To focus on H-1B applications that were most likely to enter the lottery, I first remove cases with visas other than the H-1B visa. I then exclude cases where the labor condition application was withdrawn, certified withdrawn, or denied because those cases never entered the lottery. I also remove cases from both the LCAs and the H-

1B employer data hub where the employer name contains keywords such as "university", "research", and "hospital" because these employers are cap-exempt. I consider lottery-bound H-1B applications to be between March 1, 2016, and April 3, 2016 as described in Mahajan et al. (2024), where LCAs are considered lottery-bound from March 1, 2007 to April 3, 2007. Since the 2016 H-1B lottery stopped accepting applications on April 5, 2016, I consider April 3 as the final date for likely-lottery-bound applications rather than April 5 to avoid potentially capturing endogenous application behavior at the very end of the submission window. See Figure A1 in the Appendix for a year-by-year view of weekly H-1B application timing. Finally, I restrict the sample to counties with non-zero H-1B activity, where the number of H-1B lottery applications is greater or equal to 20 in order to reduce noise.

The 2016 H-1B Lottery generated exogenous variation in the ability for firms to hire foreign high-skilled labor. A measure to estimate the win rate of H-1B visas by county, used in Mahajan et al. (2024), is the number of successful lottery applications divided by the number of likely lottery applications, defined as follows:

$$\text{Win Rate}_c = \frac{\text{Lottery Wins}_c}{\text{Likely Lottery Applications}_c}$$

where Win Rate is the H-1B lottery win rate at the county level, Lottery Wins is the data from the USCIS dataset per county, and Likely Lottery Applications is the likely lottery-bound H-1B applications from the labor condition applications dataset, also at the county level.

Aggregation of the datasets at the county level introduces some merging issues, which I address through a series of cleaning efforts. Since I first have to append a county to every zip code in the USCIS dataset using a zip-to-county crosswalk, I convert the zip codes to a character type and ensure they are all five characters in length. I repeat the same process with the LCA dataset. After cleaning, 99.9% of the zip codes in the USCIS dataset and 99.8%

of the LCA data are matched with their respective counties. To describe the distribution of my key variables, Table 1 and Appendix Figure A2 report summary statistics for my key variables. Together, this data forms the basis for evaluating how variation in the H-1B lottery affects business diversity at the county level.

Table 1: Summary Statistics for Main County-Year Variables

	Shannon Index	**HHI**	**Lottery Win Rate**
Minimum	1.376	142.9	0.00285
1st Quartile	4.426	174.9	0.20826
Median	4.518	187.3	0.31707
Mean	4.444	209.0	0.36785
3rd Quartile	4.577	205.9	0.46512
Maximum	4.736	2551.0	1.00000

Source: County Business Patterns (U.S. Census Bureau, 2025); H-1B Employer Data Hub (U.S. Citizenship and Immigration Services, 2016); LCA Disclosure Data (U.S. Department of Labor, Office of Foreign Labor Certification, 2016). Author's calculations.
Notes: This table reports summary statistics for the key variables used in the analysis. Each observation corresponds to a county-year.

6 Methodology

To estimate the effect of skilled immigration on local business diversity at the county level, I use a continuous difference-in-differences (DiD) design that leverages variation in H-1B lottery win rates across counties in 2016. I exploit the quasi-random assignment of visas in the 2016 H-1B lottery to isolate the causal impact of skilled immigration exposure on firm concentration and business diversity. This design compares changes in outcomes before and after the lottery across counties with different levels of H-1B success, while controlling for time-invariant county characteristics and common time shocks. I validate the parallel trends assumption using an event study.

6.1 Empirical Strategy

The key idea of my difference in differences specification is to compare pre- and post-lottery changes in business diversity across counties with varying levels of lottery success. My empirical specification is:

$$\log(Y_{ct}) = \beta_0 + \beta_1 \text{Win Rate}_c + \beta_2 \text{Post}_t + \beta_3 (\text{Win Rate}_c \times \text{Post}_t) + \delta_c + \gamma_t + \epsilon_{ct}$$

where Y_{ct} denotes the Herfindahl-Hirschman Index (HHI) or the Shannon Diversity Index in county c at time t, both logged to address skewness (see Figure A2 in the Appendix) and to simplify interpretations. Win Rate_c is a proxy for the proportion of successful applicants to the 2016 H-1B lottery in county c. Post_t is an indicator for years after the lottery (2017 onward). The difference in differences estimator β_3 captures the causal effect of an increase in the H-1B lottery win rate on outcome Y, the business diversity indices. All regressions include δ_c, the county fixed effects, and γ_t, the year fixed effects. Standard errors are clustered at the county level to account for serial correlation within counties over time. Estimates β_0, β_1, and β_2 are collinear with the fixed-effects and will be dropped in the estimation.

Unlike traditional DiD settings, where the treatment indicator is binary, counties in my setting are exposed to varying levels of treatment intensity, depending on how many firms located in the county won the H-1B lottery for their workers. Because the treatment is continuous rather than binary, I adopt a continuous DiD framework following the standard procedure in the literature. This approach allows me to estimate how a marginal increase in the H-1B lottery win rate affects business diversity outcomes, rather than comparing treated and untreated counties in a binary fashion. Figure 3 shows the distribution of H-1B lottery win rates across counties. Most counties fall between 0.1 and 0.6, with some bunching at the extremes. The variation present in the treatment intensity supports the use of a continuous difference-in-differences specification.

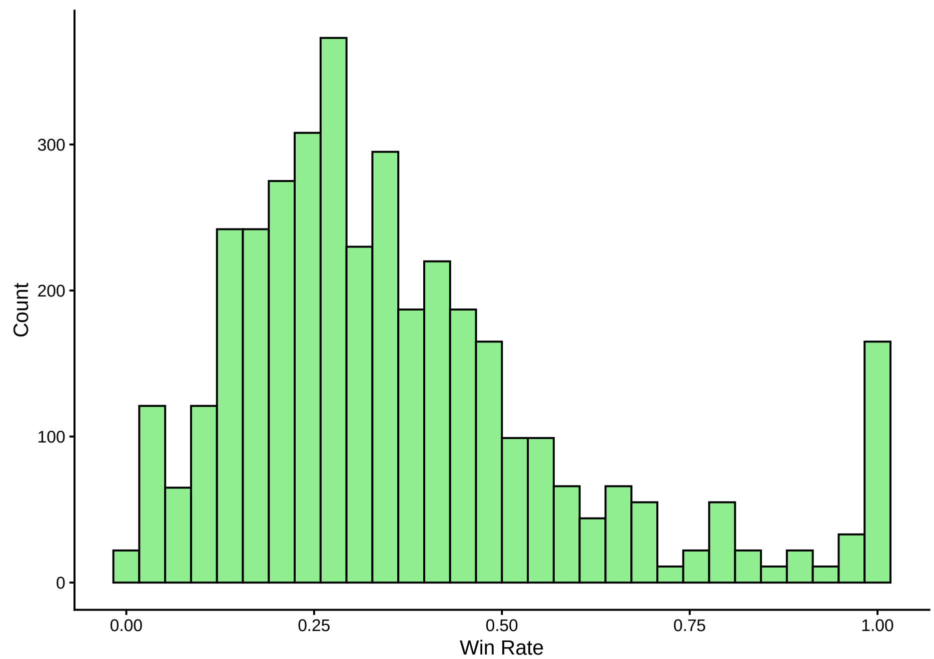

Figure 3: Distribution of H-1B Visa Win Rate. Data from LCA Disclosure Data (U.S. Department of Labor, Office of Foreign Labor Certification, 2016), H-1B Employer Data Hub (U.S. Citizenship and Immigration Services, 2016), and author's calculations.

6.2 Identification Assumptions and Validity

A key identifying assumption for this study is that the 2016 H-1B lottery generated quasi-random variation in visa allocations across counties, conditional on application. Because the H-1B lottery is randomized at the petition level, counties with more applications are more likely to experience deviations in win rates over the years due to the binomial nature of lottery outcomes. Due to randomness, some counties will overperform the national lottery success rate, while other counties will underperform. As firms are not evenly distributed in space, this randomness will aggregate into differences in H-1B inflows at the county level. These shocks are plausibly exogenous, conditional on application, and independent of unobserved determinants of business diversity at the county level, which allows for causal identification.

Another key identifying assumption for difference-in-differences is that, in the absence of treatment, the treated and control variables would have followed parallel trends over time in the outcome variable. In the continuous DiD setting, this assumption generalizes to

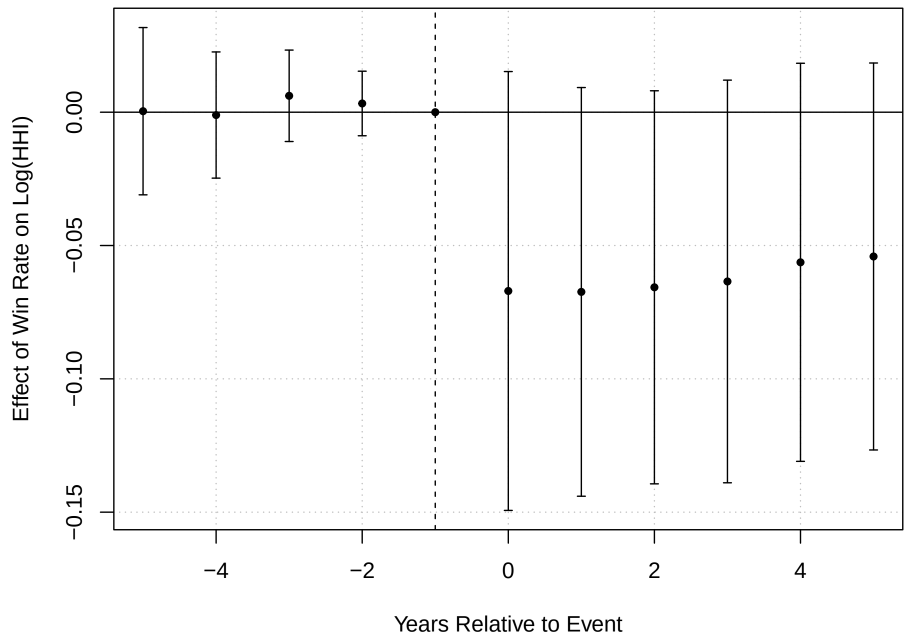

Figure 4: Event Study plot of the effect of H-1B lottery win rates on business concentration, measured as log(HHI). Error bars represent 95% confidence intervals. The omitted baseline year (t = -1) is shown at zero. Data from County Business Patterns (U.S. Census Bureau, 2025), H-1B Employer Data Hub (U.S. Citizenship and Immigration Services, 2016), LCA Disclosure Data (U.S. Department of Labor, Office of Foreign Labor Certification, 2016), and author's calculations.

requiring that counties with different levels of treatment intensity (H-1B win rates) would have exhibited parallel trends in the outcome variables (firm concentration and business diversity) in the absence of treatment. To assess the plausibility of this assumption, I conduct an event study where I interact year dummies with the continuous variable treatment. The results (Figure 4) show no significant pre-trend in the outcomes prior to 2017, suggesting that trends across counties with varying H-1B treatment intensity were parallel before treatment. In addition, a Wald test of pre-treatment interactions fails to reject joint significance ($p = 0.36$), supporting the strong parallel trends assumption discussed in Callaway, Goodman-Bacon and Sant'Anna (2024).

One limitation of using a continuous DiD design is that validating the parallel trends assumption through an event study does not guarantee identification in the presence of unobserved selection into treatment intensity. As noted by Callaway, Goodman-Bacon and

Sant'Anna (2024), even if pre-treatment outcomes evolve in parallel across treatment intensities, there may still be bias if the level of treatment applied is correlated with unobserved factors. However, in this context, treatment intensity is derived from the outcome of the 2016 H-1B lottery, which introduced exogenous variation in which firms received approvals for their H-1B visa applications. While the decision to apply may be endogenous, conditional on application, the lottery assignment introduces plausibly exogenous variation, helping to limit selection bias concerns.

Running the Callaway and Sant'anna continuous DiD estimator is not possible in my sample because there were no counties in my sample with a win-rate of zero and very few counties with near-zero win-rates. Additionally, by including zero-dose counties, I would be including counties whose firms never applied for any H-1B labor, and these counties will be systematically different than counties with H-1B activity. To address this, I instead estimate the treatment effect separately by dose quartile, reported in the robustness section below, alongside other checks.

Another limitation is that the annual nature of the H-1B lottery may violate the Stable Unit Treatment Value Assumption (SUTVA) due to counties repeatedly being exposed to the treatment across multiple years. Additionally, spatial spillovers may occur if economic activity in one county affects neighboring counties. While the 2016 H-1B lottery creates plausibly exogenous variation, counties with low win rates may still have been treated in other years, and adjacent counties may have experienced economic spillovers from treated areas. Thus, county-level spillovers may attenuate the true estimates. As such, the treatment effects should be interpreted as an average treatment effect on the treated (ATT) for counties whose treatment intensities changed due to the 2016 H-1B lottery, and not a clean one-time treatment.

7 Results

This section presents the estimated effects of the H-1B lottery on business diversity and firm concentration at the county level. I begin by estimating the average treatment effect using both the Herfindahl-Hirschman Index and the Shannon Diversity Index. I then examine firm entry as a possible mechanism using an event study and a mediation framework. Finally, I analyze heterogeneity by industry type and immigrant share to investigate whether mechanisms differ across sectors.

7.1 Main Estimates

Table 2 shows the results from the fixed-effect regressions and the continuous difference in differences model. Higher values of the Shannon index indicate greater business diversity, while lower values of the HHI indicate greater business diversity.

Table 2: Effect of H-1B Lottery Success on Business Diversity

	Pooled + Year Fixed Effects		**Difference in Differences**	
	Shannon Index	HHI	Shannon Index	HHI
Win rate × Post			**0.027***	**-0.064***
			(0.014)	(0.035)
Win rate	0.041**	-0.151**		
	(0.015)	(0.052)		
Observations	3,823	3,823	3,823	3,823
Counties	348	348	348	348
County Fixed Effects	No	No	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
R^2	0.06	0.06	0.81	0.87

Source: County Business Patterns (U.S. Census Bureau, 2025); H-1B Employer Data Hub (U.S. Citizenship and Immigration Services, 2016); LCA Disclosure Data (U.S. Department of Labor, Office of Foreign Labor Certification, 2016). Author's calculations.
Notes: * p < 0.10, ** p < 0.05, *** p < 0.01. Standard errors are clustered by county. The Shannon Index and HHI labels are measured through log(Shannon Index) and log(HHI). Win rate is measured from 0 to 1. For interpretation, a 10 percentage point increase corresponds to a 0.001 unit change in win rate. Three counties have no measured win rate and are dropped from every column, so the estimation sample is 348 of the 351 counties in the panel.

In the fixed effect regressions, a 10 percentage point increase in H-1B lottery win rates in a county is associated with an estimated 1.5% decrease ($e^{-.151}$) in the Herfindahl-Hirschman Index and a 0.4% increase in the Shannon index. County fixed effects were excluded from the cross-sectional regressions in columns 1 and 2 because win rate is defined at the county level and does not vary over time. Including both county and year fixed effects would result in perfect multicollinearity. These estimates reflect cross-sectional associations between counties, suggesting that counties with higher H-1B win rates tend to exhibit greater business diversity and less market concentration than those with lower H-1B win rates. One possible interpretation is that such counties may already have a more dynamic firm ecosystem, greater agglomeration effects, or a higher baseline demand for skilled labor, which increases both H-1B applications and business diversity. In addition, the coefficient on the interaction term `win rate` $\times$ `post`, which captures the causal effect of an increase in the H-1B lottery win rate, is positive for the Shannon index and negative for HHI.

In the DiD specification, a 10 percentage point increase in lottery win rates is associated with a 0.27% increase in the Shannon index ($p < 0.10$) and a 0.64% decrease in HHI ($p < 0.10$). These estimates correspond to approximately 25% of a standard deviation in log(HHI) and 31% of a standard deviation in log(Shannon) across counties, suggesting that skilled immigration not only modestly increases business diversity but meaningfully reduces firm concentration. To put this effect into context, 25% of a standard deviation in HHI, a magnitude of about 32, corresponds roughly to the difference between a county at the median and a county at the 55th percentile for HHI in the 2022 County Business Patterns dataset. For example, this shift is similar to the difference in HHI between Miami-Dade County, Florida—a large immigrant-rich metro with a broad mix of service sectors, and St. Louis County, Missouri—which has a more concentrated economic base anchored around healthcare and education. These effects are consistent with the hypothesis that H-1B workers contribute to a broader distribution of firm types, particularly through spillover effects, entrepreneurship, or indirect labor market effects such as spouses of H-1B workers starting small businesses.

The baseline associations of `win rate` are statistically significant across all models, capturing persistent cross-sectional differences that predate the lottery. While informative, these associations should not be interpreted as causal, and only the interaction term `win rate` × `post` should be interpreted as causal. In addition, since the H-1B win rate is a proxy constructed from administrative records and filtered applications, classical measurement error may inflate the denominator and attenuate the estimated coefficients towards zero. Thus, the true causal estimate may be larger than reported.

7.2 Mechanisms: Firm Entry

A possible mechanism for the observed increase in business diversity following a rise in the H-1B win rate is an increase in new business formation after the 2016 H-1B lottery. To test this mechanism, I conduct an event study (Figure 5) examining the effect of the H-1B lottery win rate on the log of establishment counts at the county level. The event study supports the validity of the parallel trends assumption prior to the treatment year (2016) and reveals a clear upward divergence in establishment counts after the lottery. This pattern is consistent with a causal interpretation that H-1B lottery success resulted in new business formation.

To formally test whether business formation mediates the effect of the H-1B lottery success on market concentration, I estimate the following three regressions: The first stage regression is the original continuous differences in differences regression, which estimates the total effect of the H-1B lottery on market concentration:

$$\log(\text{HHI}_{ct}) = \beta \text{Win Rate}_c \times \text{Post}_t + \delta_c + \gamma_t + \epsilon_{ct}$$

The second stage regression applies the same DiD framework, but uses the log of establishment counts in each county-year as the dependent variable to measure the causal effect

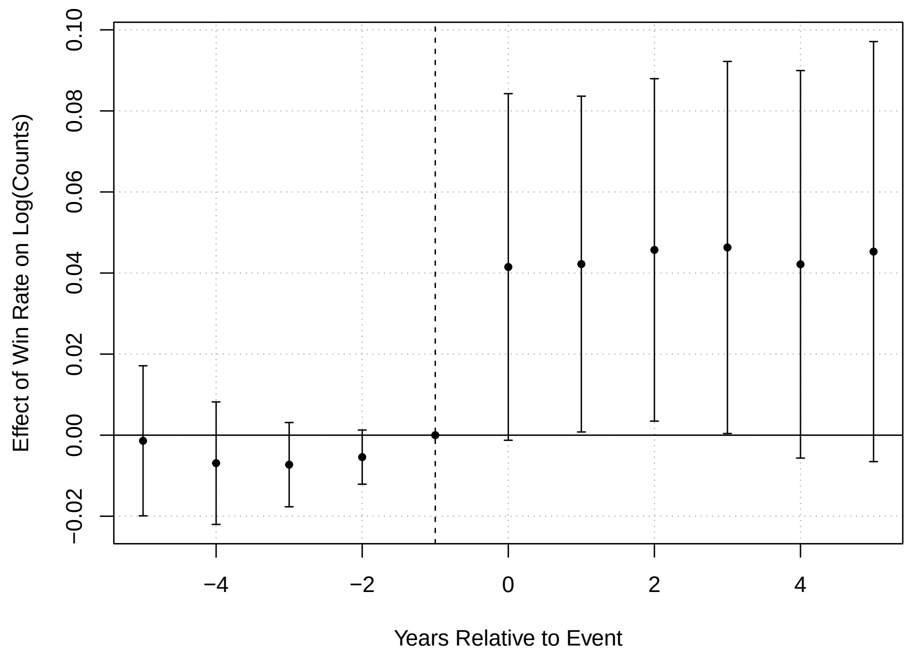

Figure 5: Event Study plot of the effect of H-1B lottery win rates on the number of businesses, measured as log(Counts). Error bars represent 95% confidence intervals. Data from County Business Patterns (U.S. Census Bureau, 2025), H-1B Employer Data Hub (U.S. Citizenship and Immigration Services, 2016), LCA Disclosure Data (U.S. Department of Labor, Office of Foreign Labor Certification, 2016), and author's calculations.

of the H-1B lottery on business formation.

$$\log(\text{Establishment Counts}_{ct}) = \beta \text{Win Rate}_c \times \text{Post}_t + \delta_c + \gamma_t + \epsilon_{ct}$$

Finally, the mediation regression adds log(Establishment Counts) as a control for the original DiD regression. If business formation is a key mechanism, I expect the coefficient on Win Rate × Post to attenuate or become statistically insignificant as log(Establishment Counts) will have absorbed all the variation.

$$\log(\text{HHI}_{ct}) = \beta_3 \text{Win Rate}_c \times \text{Post}_t + \beta_4 \log(\text{Establishment Counts}) + \delta_c + \gamma_t + \epsilon_{ct}$$

Table 3 shows the coefficients for the three regressions composing of my mediation strategy. The coefficient on log(Establishment Counts) suggests that a 1 percent increase in estab-

Table 3: Mechanism Analysis: Business Formation and Market Concentration

	HHI	**Establishment Counts**	**HHI w/ Mediator**
Win Rate × Post	-0.066*	0.050**	-0.022
	(0.035)	(0.026)	(0.023)
log(Establishment Counts)			-0.865***
			(0.151)
Observations	3,823	3,823	3,823
County Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes

Source: County Business Patterns (U.S. Census Bureau, 2025); H-1B Employer Data Hub (U.S. Citizenship and Immigration Services, 2016); LCA Disclosure Data (U.S. Department of Labor, Office of Foreign Labor Certification, 2016). Author's calculations.

Notes: Standard errors clustered by county in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Column (1) estimates the total effect of H-1B win rate on the Herfindahl-Hirschman Index (main regression result). Column (2) estimates the effect of H-1B win rate on business formation. Column (3) estimates the regression in (1) while controlling for log(Establishment Counts) to examine whether increases in establishment counts mediate the observed reduction in concentration.

lishment counts is associated with a 0.87 percent decrease in HHI. As expected, the DiD estimator, Win Rate × Post becomes statistically insignificant once log(Establishment Counts) is included as a control, consistent with the hypothesis that business formation mediates the relationship between H-1B lottery success and market concentration. Notably, a one percentage point increase in the lottery win rate leads to a 5% increase in establishment counts, which suggests a modest amount of new business formation when counties are exposed to skilled immigration.

The attenuation of the Win Rate × Post coefficient from -0.066 to -0.022, and its loss of statistical significance, suggests that roughly two-thirds of the total effect of exposure to the H-1B lottery to business diversity operates through increased business formation. This points to new firm entry as a substantial component of the effect of the H-1B lottery on business diversity and on the broader economic impact of skilled immigration. These findings strongly suggest that skilled immigration may foster dynamic firm ecosystems through entrepreneurial spillovers. While this strategy provides suggestive evidence of a mediation channel, it does not satisfy the strict identification assumptions required for formal causal mediation analysis.

7.3 Heterogeneity

To examine heterogenous effects of the H-1B lottery by industry type, I modify the main specification to operate at the county-industry-year level. To investigate whether certain sectors are more sensitive to H-1B inflows, I estimate the effect separately for high-skill and low-skill immigrant-intensive industries using the following specification:

$$\log(\text{Establishment Counts}_{ctn}) = \beta \text{Win Rate}_c \times \text{Post}_t + \delta_c + \gamma_t + \rho_n + \epsilon_{ctn}$$

where c, t, and n represent counties, years, and four-digit NAICS industries, respectively. Since I use industry-level data to analyze heterogeneity, the dependent variables can no longer be the diversity indices because they were generated through the aggregation of the NAICS categories by county. Therefore, I use establishment counts as my dependent variable. The model includes county (δ_c), year (γ_t), and four-digit NAICS category (ρ_n) fixed effects.

To classify immigrant-intensive industries, I draw on academic literature and data from the US Census Bureau's Annual Business Survey (U.S. Census Bureau, 2022), which documents high-rates of immigrant business ownership in specific sectors. Following Fairlie and Lofstrom (2015), Kerr and Mandorff (2021), and Kerr and Lincoln (2010), as well as data from the US Census Bureau, I define immigrant-intensive sectors to include the industries listed in Appendix Table A1. I then estimate the same specification above for all immigrant-intensive industries, only high-skilled industries, and only low-skilled industries, and the results are in Table 4. NAICS sectors are defined as "high-skilled" if they are predominantly staffed by workers requiring a bachelor's degree or above or involve non-routine cognitive tasks (Autor, Levy and Murnane, 2003). While the list of immigrant-intensive industries draws on existing literature and data sources, some classification decisions involved researcher judgement based on qualitative descriptions and observed immigrant ownership patterns.

The interpretation of the results for all sectors is that a 1 percentage point increase

Table 4: Effect of H-1B Lottery Success on Establishment Counts in Immigrant-Intensive Industries

	All Sectors	**Immigrant-Intensive**	**High-Skill**	**Low-Skill**
Win rate × Post	0.0022***	0.0058***	0.0082***	0.0047**
	(0.0008)	(0.0017)	(0.0027)	(0.0013)
Observations	764,950	113,128	32,332	80,796
County Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Industry Fixed Effects	Yes	Yes	Yes	Yes

Source: County Business Patterns (U.S. Census Bureau, 2025); H-1B Employer Data Hub (U.S. Citizenship and Immigration Services, 2016); LCA Disclosure Data (U.S. Department of Labor, Office of Foreign Labor Certification, 2016); Annual Business Survey (U.S. Census Bureau, 2022). Author's calculations. *Notes:* Standard errors clustered by county in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Columns (3), High-Skill and (4), Low-Skill are split from the immigrant-intensive industries. Industry classifications follow Appendix Table A1.

in the H-1B lottery win rate at the county level results in a 0.22 percent increase in the number of businesses across all NAICS categories since the dependent variable is logged. This is the same as saying a 10 percentage point increase results in a 2 percent increase in the number of firms. Because the effect is statistically significant, it is likely that firm entry occurs broadly, contributing to business diversity at the county level. However, the effect size for the immigrant-intensive industries is nearly three times that of the effect size across all sectors. In addition, the effect size for high-skill immigrant-intensive firms is nearly double the effect size for low-skill immigrant-intensive firms, strongly suggesting that H-1B labor allows firms to increase productivity and expansion, as well as suggesting that H-1B visa holders start firms themselves. This aligns with the previous literature, such as Mahajan et al. (2024), Kerr, Kerr and Lincoln (2015), and Doran, Gelber and Isen (2022), which document the productivity gains from firms that hire H-1B workers.

The modest but statistically significant increase in establishment counts for low-skilled immigrant-intensive industries suggests a different mechanism from the one driving the growth in high-skilled sectors. Since H-1B visa holders are generally restricted to high-skilled employment in sponsoring firms, the expansion of low-skilled firms such as restaurants, salons, or other service-sector establishments is likely driven by spousal entrepreneurship or community spillovers. Many H-1B visas are accompanied by spouses on H-4 visas, which usually does not allow employment at high-skilled firms. In addition, the presence of high-skilled immigrants in a neighborhood can generate demand-side spillovers. These indirect pathways can explain why firm formation increased in sectors not traditionally associated with the H-1B visa.

7.4 Robustness Checks

The main results are robust to a variety of alternative specifications. First, I replicated my analysis with both the Shannon Diversity Index and Herfindahl-Hirschman Index to measure business diversity as shown in my main results table as shown in Table 2. The estimated effects are consistent across both metrics in magnitude and significance. Second, I re-estimated the model excluding years after the start of the pandemic (2020 and onwards) to account for abnormal business closures. Throughout this, the results remain unchanged. I conducted several placebo tests by reassigning lottery treatment to earlier years. While placebo effects are insignificant for 2014 and earlier, there are statistically significant effects in 2015 and 2016.

The years of 2015 and 2016 follow earlier H-1B lotteries in 2014 and 2015, meaning that counties may have already experienced treatment effects from prior visa allocations. As such, 2015 and 2016 are not valid untreated counterfactuals. This reflects a potential violation of the Stable Unit Treatment Value Assumption (SUTVA), specifically the assumption that units (counties) are fully treated or untreated. Therefore, I rely on 2014 and earlier as cleaner placebo years to assess robustness. While prior H-1B lotteries may have affected outcomes in some counties before 2017, my estimates should be interpreted as a dose-response effect of increased H-1B inflows from the 2016 H-1B lottery. This interpretation is consistent with a limited SUTVA violation, in which units may have received prior treatments. This interpretation still remains valid for estimating incremental effects under continuous treatment intensity.

Table 5: Robustness Checks for DiD Specification

Specification	Coefficient	Std. Error	Significance
Placebo (FY2014)	-0.040	0.026	
Placebo (FY2015)	-0.049	0.028	*
Placebo (FY2016)	-0.057	0.031	*
Excl. COVID Years	-0.070	0.039	*
Restrict to Local	-0.004	0.001	**
Restrict to Tradable	-0.004	0.004	
Lags (1 and 2 year)	-0.032	0.018	*
Main (Driscoll-Kraay SE)	-0.066	1.53e-9	***
Placebo (Driscoll-Kraay SE)	-0.057	0.009	***

Source: County Business Patterns (U.S. Census Bureau, 2025); H-1B Employer Data Hub (U.S. Citizenship and Immigration Services, 2016); LCA Disclosure Data (U.S. Department of Labor, Office of Foreign Labor Certification, 2016). Author's calculations.
Notes: This table reports the coefficient on the interaction term `win rate` × `post` (or `post_placebo`) across several robustness specifications. Standard errors for the Driscoll-Kraay models are robust to heteroskedasticity, autocorrelation, and cross-sectional dependence. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Next, I re-estimated my main DiD model using data restricted to local industries (NAICS codes: 44, 45, 62, 71, 72, 81). After restricting the data to local industries, the results remain negative and significant, consistent with the main regressions. In addition, restricting the data to tradable industries resulted in insignificant results, meaning my main results are likely not driven by SUTVA violations. Together, these robustness checks support the validity of the identification strategy and reinforce the causal interpretation. In addition, I also include one- and two-year lags to account for persistence. The one-year lag is significant, with a coefficient of 0.61, while the 2-year lag is negligible. The treatment effect remains significant, with a coefficient of -0.032, consistent with gradual adjustment mechanisms such as obtaining green cards, spousal work permits, or startup creation.

Finally, I re-estimate both the main and placebo regressions using Driscoll-Kraay standard errors to account for heteroskedasticity, autocorrelation, and cross-sectional dependence. While this approach yields highly significant p-values for both specifications, the estimated placebo effect is substantially smaller in magnitude and precision than the main effect. However, when I apply Driscoll-Kraay standard errors to the event study regression, I find that the pre-treatment coefficients, which were not statistically significant under county-clustered standard errors, became spuriously significant, which is shown in Appendix Figure A3. This suggests that the Driscoll-Kraay standard errors may overstate significance in this context, and therefore, I rely primarily on county-clustering standard errors for inference. Taken together, these checks support the identification strategy, subject to the sensitivity to the sample definition documented in the next subsection.

7.5 Dose-Response Heterogeneity

As discussed in the identification section, the estimator of Callaway, Goodman-Bacon and Sant'Anna (2024) cannot be implemented because it estimates the average causal response at dose d by comparing units at that dose against untreated units, and no county in the sample has a win rate of zero. The substantive concern that motivates their estimator nonetheless applies. With a continuous treatment, the coefficient on Win Rate × Post is a weighted average of heterogeneous causal responses across the dose distribution. This implies that weights may be different than what the researcher chooses. By binning my estimate into quartiles, this directly tests it while using only variation within the sample of lottery applicants, which preserves the randomization that identifies the main estimate.

I sort counties into quartiles of the win-rate distribution and estimate

$$\log(Y_{ct}) = \sum_{q=2}^{4} \beta_q \left(\mathbf{1}[Q_c = q] \times \text{Post}_t\right) + \delta_c + \gamma_t + \epsilon_{ct}$$

where Q_c is county c's dose quartile. Quartiles are cut on the county-level distribution, since the win rate is time-invariant and cutting the stacked panel would weigh each county by its number of observed years. County and Year fixed effects are included.

The mean win rate rises from 0.13 in the lowest quartile to 0.26, 0.38, and 0.69 in the second, third, and fourth. The estimated effects on log(HHI) are -0.008 (standard error of 0.032) for the second quartile, -0.037 (0.019) for the third, and -0.041 (0.022) for the fourth. Full estimates, including the Shannon results and the linear specification estimated on the same sample, are reported in Appendix Table A2. The event study for quartiles is consistent with the parallel trends evidence reported above: none of the twelve pre-treatment interactions is significant at the five percent level, and only one, the third quartile two years before the lottery, reaches significance at the ten percent level (Appendix Figure A4).

The response is therefore not proportional to dose. If it were linear, the top quartile, with a mean dose of 0.69, should show roughly twice the effect of the third, with a mean dose of 0.38. Instead the two are statistically indistinguishable: their difference is 0.004 with a standard error of 0.016 ($p = 0.78$). Figure 6 plots the estimated coefficients against mean dose. The effect appears concentrated between the second and third quartiles and flat above.

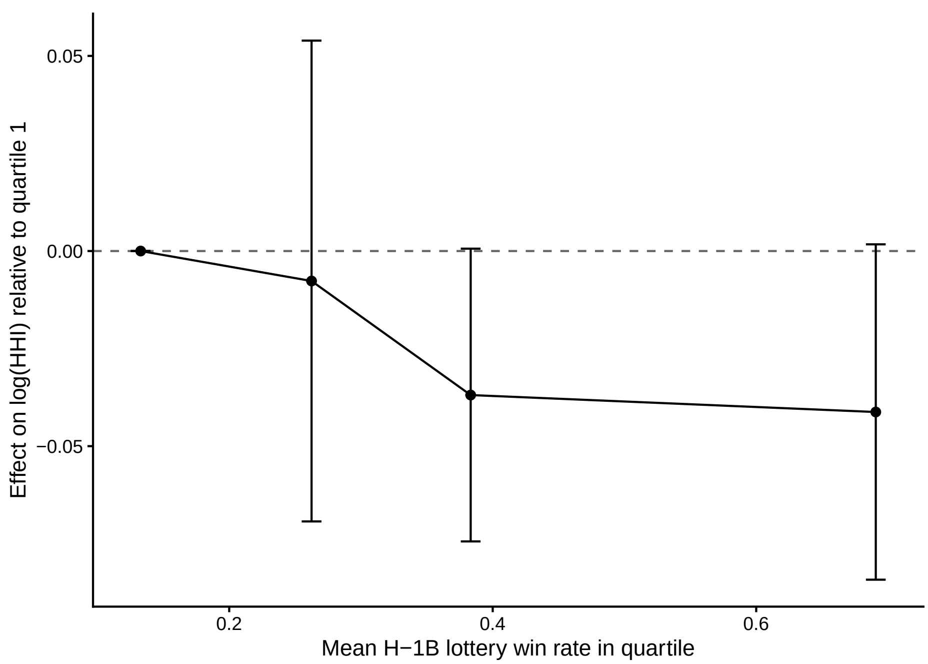

Figure 6: Estimated effect on log(HHI) by dose quartile, plotted against the mean win rate within each quartile. Standard errors are clustered by county. The lowest quartile is the omitted reference category and is normalized to zero. Data from author's calculations.

However, this cannot be interpreted as evidence of a threshold effect, because it does not survive scrutiny. Dropping the fourteen counties whose win rate is censored at one leaves it intact, with the third-quartile coefficient at -0.046. But restricting the sample to counties with at least 100 LCA positions removes it: the third-quartile estimate falls to −0.015 and the fourth changes sign to +0.019. The explanation of this behavior is that median petitions fall monotonically across dose quartiles, from 208 in the lowest to 198, 91, and 45 in the highest. Therefore, the upper bins are disproportionately built from counties whose win rate rests on a small denominator.

7.6 Sensitivity to the Petition Threshold

The main sample was computed with a threshold where the number of H-1B lottery applications is greater than or equal to 20. As another robustness check, I change the threshold to test the sensitivity of my estimates.

Figure 7 shows a sweep of the minimum from one petition to 50, while full point estimates and standard errors are reported in Appendix Table A3. The estimate is not monotone while changing the threshold. For log(HHI) it runs +0.032 at a floor of one, −0.052 at ten, −0.064 at twenty, −0.058 at thirty, −0.013 at forty, and −0.004 at fifty; the Shannon estimates mirror this pattern with the opposite sign throughout. The estimate is largest at the threshold used in the main specification and attenuates in both directions. Appendix Figure A5 shows how low numbers of petitions causes censoring at 0 and 1.

There are three possible explanations. The first is classical measurement error: win rates built on few petitions are noisy, which attenuates the coefficient toward zero. This can be assessed directly. Treating each county's win rate as a binomial draw with sampling variance $\hat{w}(1-\hat{w})/n$, the implied reliability of the dose — the share of observed cross-county variance that is not sampling noise — is 0.95 at the twenty-petition threshold and 0.85 even at a threshold of one. Correcting the main estimate for classical attenuation moves it only from −0.064 to −0.068. Sampling noise is therefore too small to generate the observed sensitivity, and in any case attenuation implies that raising the threshold should strengthen the estimate rather than eliminate it.

A second possibility is that raising the threshold selects larger, more urban counties, such

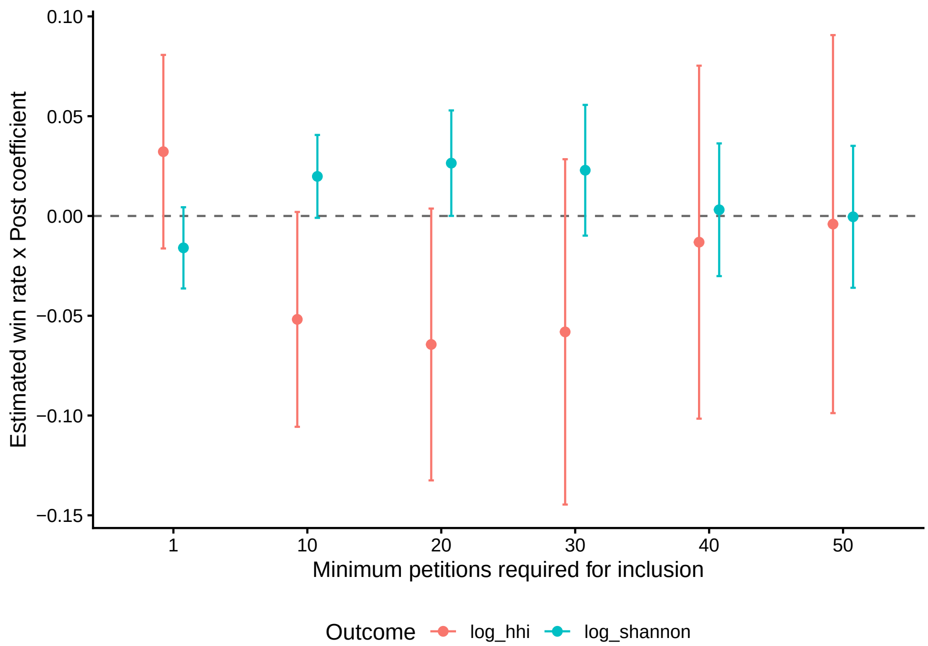

Figure 7: Estimated coefficient on Win Rate × Post as the minimum-petition threshold varies, with 95 percent confidence intervals from county-clustered standard errors. The vertical axis is the estimated coefficient; the horizontal axis is the minimum number of LCA positions required for a county to enter the sample. Author's calculations.

that the change reflects a shifting population rather than a cleaner measure. The composition data limit how much of the movement this can explain. Median establishments per county rise from around 2,000 at a floor of one to around 9,300 at a floor of 20, so the low end of the sweep does shift the sample substantially. But the estimate has already collapsed by a floor of 40, where the median is around 12,600, a 36 percent shift from the floor-of-20 sample against the 4.7-fold shift between floors of one and 20, and it moves only another 2 percent by a floor of 50. The result therefore weakens across a range where sample composition barely moves, which is difficult to attribute to urban selection alone. Thresholds above 50 would separate the two explanations less cleanly rather than more, since they select on county size heavily enough that any movement is as easily read as urban heterogeneity, and they are not reported.

A third possibility, which is not resolved here, is that the win rate captures more than lottery luck and may capture measurement issues that have already been noted. Mainly, imperfect identification of cap-exempt employers, a numerator built on the public H-1B employer datahub, rather than all firms with H-1B labor that won the lottery, and a denominator built from LCA positions rather than lottery registrations. All of this may vary systematically with local industry composition.

This limitation is common to the H-1B lottery literature broadly. Even papers that have FOIA access to USCIS petition-level data, such as Mahajan et al. (2024), still rely on LCA filings to proxy the denominator of likely lottery applications, since FOIA identifies the actual lottery winners but not the universe of lottery entrants. This study lacks the EIN-level linkage that FOIA provides, instead relying on zip-code matches and keyword-based filtering on the numerator side as well, which may introduce additional imprecision. I therefore report the main estimate as conditional on the sample definition rather than invariant to it, and read the magnitude as suggestive rather than precisely identified.

7.7 Discussion

The main results suggest that H-1B visa lottery success increases business diversity and reduces firm concentration at the county level. This effect is modest but statistically significant and consistent across the Shannon Index and the Herfindahl-Hirschman Index. A key mechanism appears to be new business formation, potentially from productivity spillovers, agglomeration effects, and immigrant entrepreneurship in high-skilled industries. Other channels include spousal entrepreneurship or community spillovers, all of which can enhance the resilience of local business ecosystems. The heterogeneous effects suggest that new firm entry occurred over a broad range of NAICS categories in both high-skilled and low-skilled immigrant-intensive industries, consistent with the decline in HHI. While this study formally tests business formation as a mechanism, other potential channels, such as productivity spillovers and agglomeration, are not tested causally and warrant further investigation.

While these effects align with prior literature on the positive firm-level or city-level effects of skilled immigration, this paper is among the first to identify corresponding changes in business diversity at the county level. The use of both the Shannon Index and HHI, as well as the robustness tests conducted, strengthens the internal validity of this finding. However, several limitations still remain. The magnitude of the estimate is sensitive to the minimum-petition threshold that defines the sample, and that sensitivity is not fully explained by either sampling noise in the win rate or by the shift toward larger counties that a higher threshold induces. The analysis relies on one year of lottery variations (2016), and county-level aggregation may obscure firm-level heterogeneity. Additionally, while the H-1B lottery introduces exogenous variation, the decision to apply is still endogenous; firms self-select into applying for the H-1B lottery. This may introduce residual selection bias, though this issue is common to all H-1B lottery-based studies due to the nature of the H-1B visa. Therefore, since lottery applications are random, conditional on application, my identification strategy identifies the marginal effect of winning the lottery, not the full effect of applying for H-1B workers. Despite these limitations, this study provides novel evidence that skilled immigration not only shapes firm performance but also the structure and diversity of local business environments.

8 Conclusion

High-skilled immigration through the H-1B lottery improves business diversity, increases the number of businesses, and reduces firm concentration at the county level. Using a continuous difference in differences framework, I identify causal evidence that higher H-1B lottery win rates lead to modest, but statistically significant, increases in business diversity and reductions in firm concentration. These results are robust across two different measures of business diversity, and hold under several alternative specifications, including placebo tests, exclusions of post-COVID years, and local and tradable industry restrictions. These findings contribute to a growing body of research documenting the economic effects of high-skilled immigration. While prior work has focused on employment, wages, or innovation at the firm or city level, this paper is among the first to identify structural changes in local business ecosystems.

Several limitations remain. The analysis uses only one year of lottery variations and is limited to county-level outcomes, which may obscure important firm-level heterogeneity. Furthermore, while the H-1B lottery itself creates plausibly exogenous variation, firms self-select into the application process, potentially introducing selection bias. Nonetheless, the result is consistent with a growing body of research that suggests high-skilled immigration contributes to not only firm-level performance, productivity growth, and increased agglomeration effects, but also to more dynamic local economies. This study's findings suggest that skilled immigration contributes to a more dynamic and diverse business environment. As policymakers consider reforms to the H-1B program, these broader economic impacts deserve careful consideration alongside labor market concerns.

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Appendix A: Additional Figures and Tables

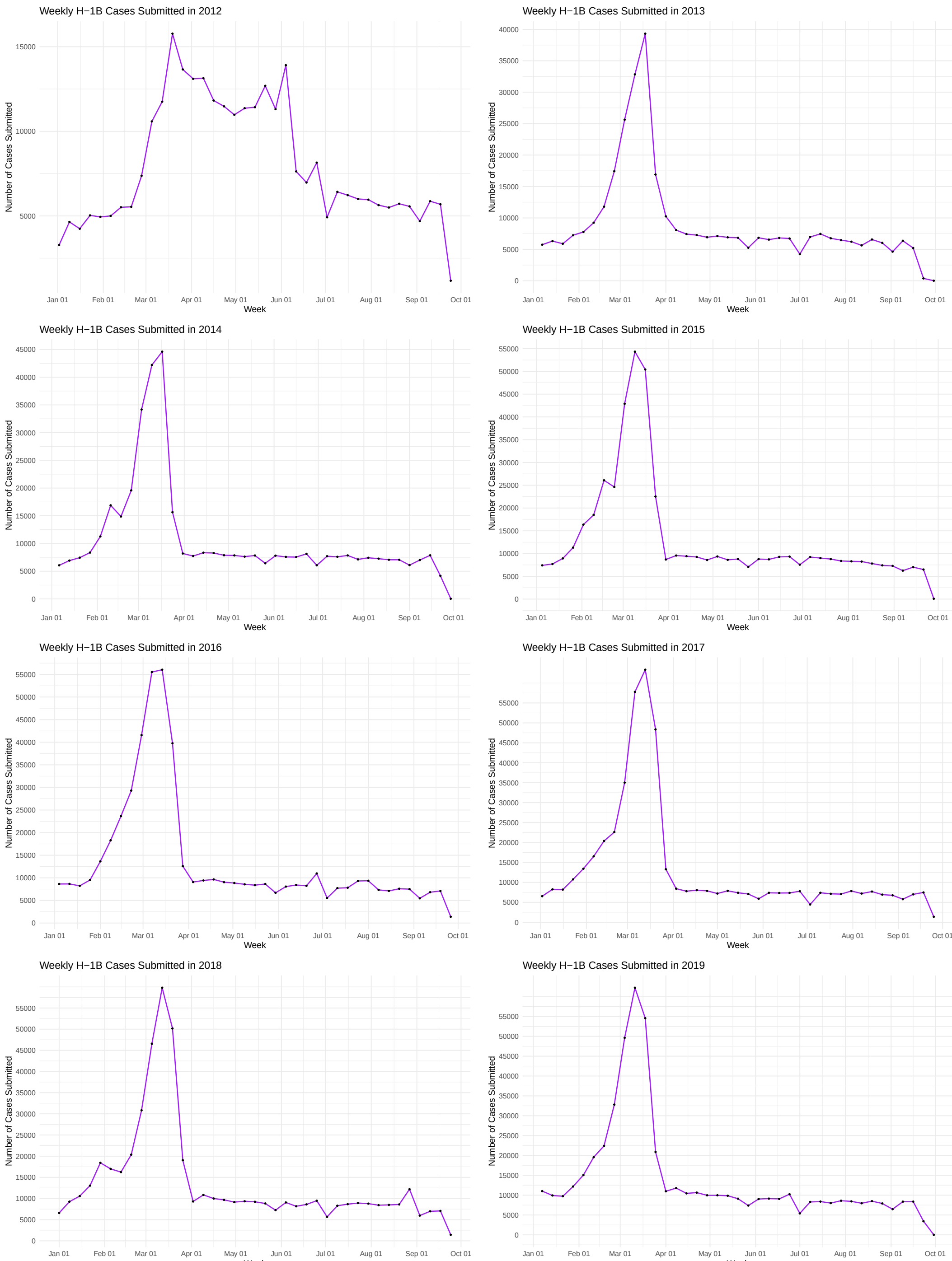

Figure A1: Weekly H-1B Visa Submissions by Year (2012–2019). Each panel shows a sharp spike in case submissions leading up to the annual lottery.

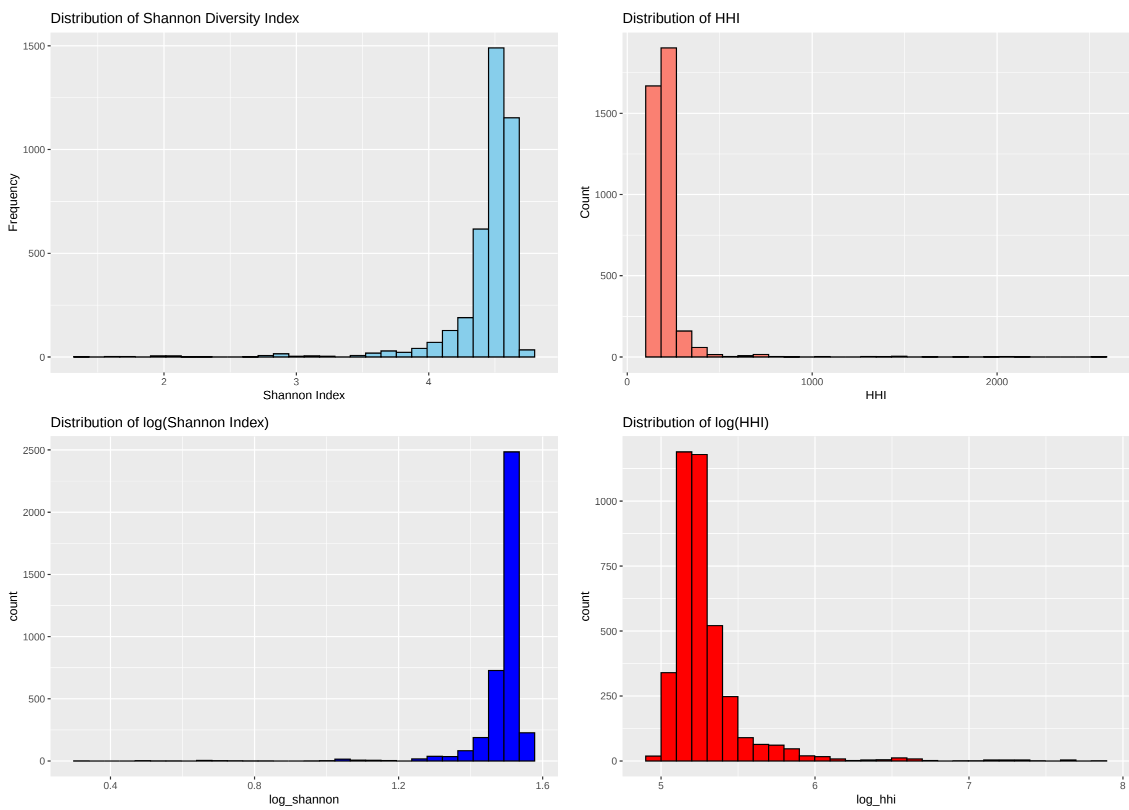

Figure A2: Distribution of Shannon Index and HHI vs The Distribution of Log Shannon and Log HHI

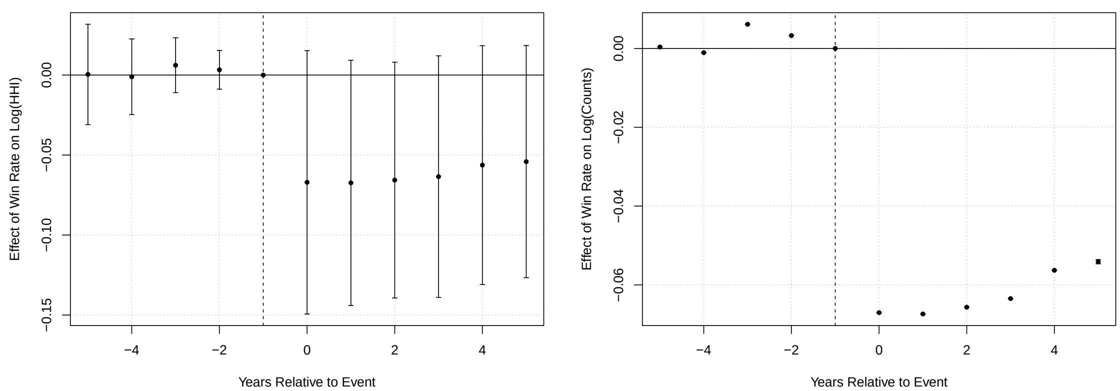

Figure A3: Comparison Between Event Study Using County-Clustered Standard Errors (Left) and Driscoll Kray Standard Errors (Right)

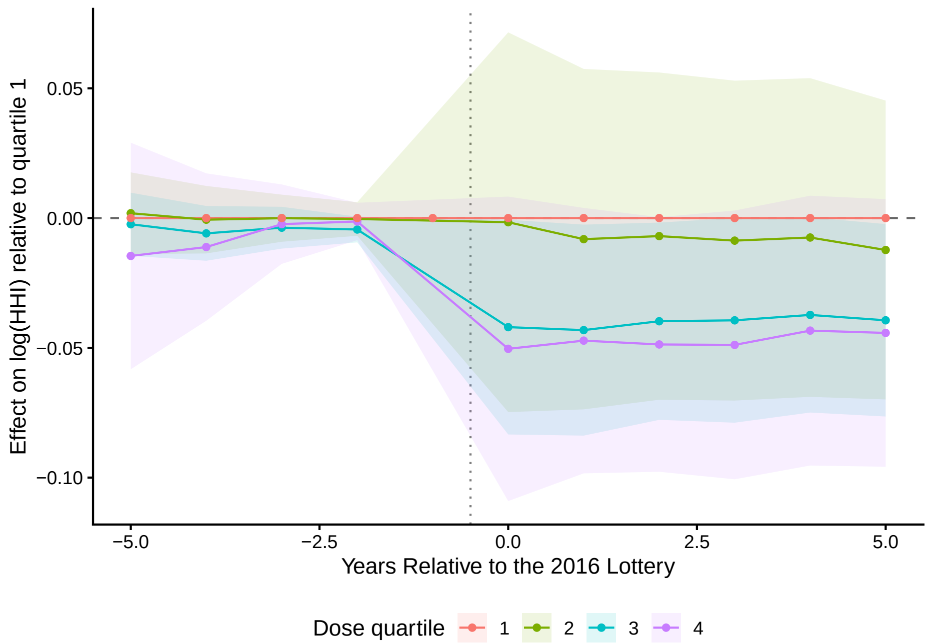

Figure A4: Event Study on log(HHI) by Dose Quartile. 95 percent confidence intervals plotted from county-clustered standard errors. Data from author's calculations.

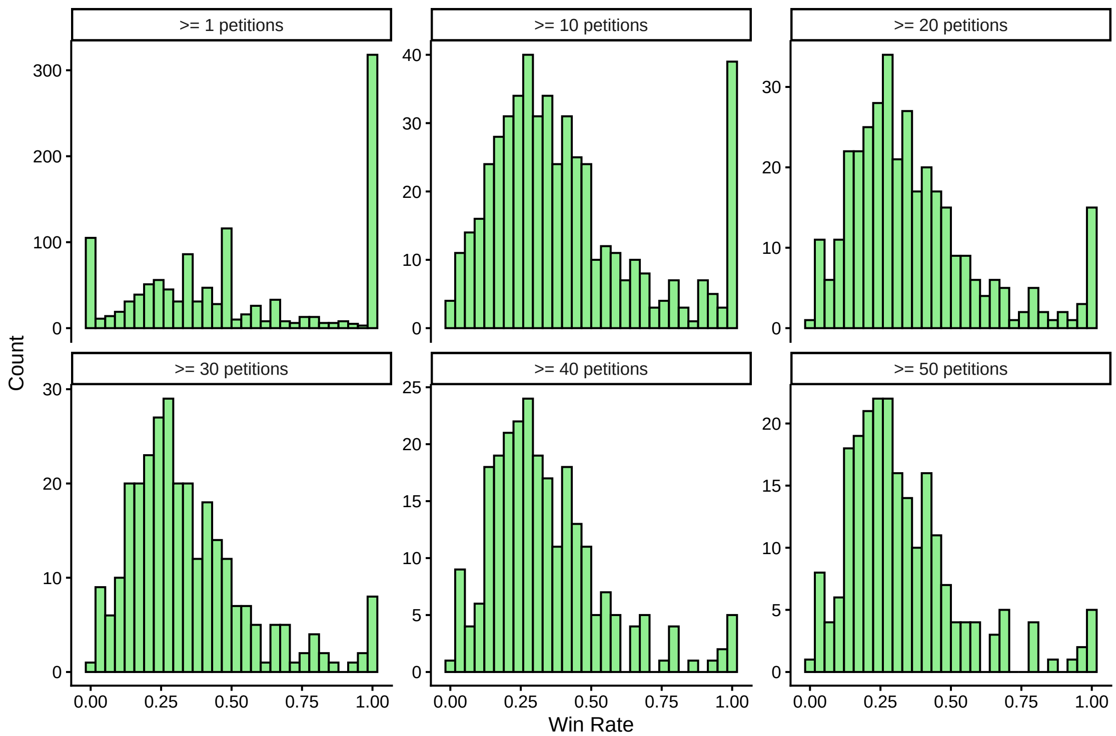

Figure A5: Win Rate Distribution by Floor of the Number of Petitions

Table A1: Immigrant-Intensive Industries by NAICS Type

NAICS	Skill Type	Industry Description
5112	High	Software Publishers
5182	High	Data Processing, Hosting, and Related Services
5191	High	Other Information Services
5242	High	Insurance Agencies, Brokerages, and Related Services
5412	High	Accounting, Tax Preparation, Bookkeeping
5413	High	Architectural, Engineering, and Related Services
5415	High	Computer Systems Design and Related Services
5416	High	Management, Scientific, and Technical Consulting
6211	High	Offices of Physicians
2382	Low	Building Equipment Contractors (e.g., electrical, HVAC)
2383	Low	Building Finishing Contractors (e.g., drywall, flooring)
2389	Low	Other Specialty Trade Contractors
3118	Low	Bakeries and Tortilla Manufacturing
4451	Low	Grocery Stores (e.g., corner and ethnic stores)
4461	Low	Health and Personal Care Stores (e.g., pharmacies)
4471	Low	Gasoline Stations
4481	Low	Clothing Stores
4853	Low	Taxi and Limousine Services
5613	Low	Employment Services
5617	Low	Services to Buildings and Dwellings
6216	Low	Home Health Care Services
6231	Low	Nursing Care Facilities
6232	Low	Residential Intellectual and Developmental Disability Facilities
6233	Low	Continuing Care Retirement Communities and Assisted Living
6241	Low	Individual and Family Services
6244	Low	Child Day Care Services
7211	Low	Hotels and Motels (except Casino Hotels)
7225	Low	Restaurants and Other Eating Places
8111	Low	Automotive Repair and Maintenance
8121	Low	Personal Care Services (e.g., salons, barbershops)
8123	Low	Drycleaning and Laundry Services

Source: Annual Business Survey (U.S. Census Bureau, 2022); Fairlie and Lofstrom (2015); Kerr and Mandorff (2021); Kerr and Lincoln (2010).
Notes: 4-digit NAICS categories classified as immigrant-intensive, split into high-skill and low-skill groups. These classifications define the subsamples used in Table 4.

Table A2: Dose-Response Estimates by Win-Rate Quartile

	(1)	(2)	(3)
Dependent Variable	log(HHI)	log(Shannon)	log(HHI)
Quartile 2 × Post	-0.0078	0.0026	
	(0.0315)	(0.0129)	
Quartile 3 × Post	-0.0371*	0.0176**	
	(0.0193)	(0.0072)	
Quartile 4 × Post	-0.0414*	0.0157**	
	(0.0221)	(0.0079)	
Win Rate × Post			-0.0644*
			(0.0348)
County fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Counties	348	348	348
Observations	3,823	3,823	3,823
R^2	0.868	0.815	0.868
Within R^2	0.010	0.013	0.007

Source: County Business Patterns (U.S. Census Bureau, 2025); H-1B Employer Data Hub (U.S. Citizenship and Immigration Services, 2016); LCA Disclosure Data (U.S. Department of Labor, Office of Foreign Labor Certification, 2016). Author's calculations.

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors clustered by county in parentheses. Columns (1) and (2) sort counties into quartiles of the county-level win-rate distribution and interact each quartile indicator with `Post`. The lowest quartile is the omitted reference category, so all coefficients are relative to the lowest quartile. Each quartile contains 87 counties, with mean win rates of 0.134, 0.263, 0.383, and 0.691 from the lowest quartile to the highest. Column (3) reports the linear continuous-dose specification estimated on the same sample for comparison.

Table A3: Sensitivity of the Continuous DiD Estimate to the Minimum-Petition Threshold

	(1)	(2)	(3)	(4)	(5)	(6)
Minimum LCA positions	1	10	20	30	40	50
Win Rate × Post	0.0322	-0.0518*	-0.0644*	-0.0581	-0.0131	-0.0041
	(0.0247)	(0.0275)	(0.0347)	(0.0441)	(0.0451)	(0.0483)
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Counties	1,189	501	348	292	253	228
County-years	13,063	5,504	3,823	3,207	2,778	2,503
Median LCA positions	6	40	97	167	230	294

Source: County Business Patterns (U.S. Census Bureau, 2025); H-1B Employer Data Hub (U.S. Citizenship and Immigration Services, 2016); LCA Disclosure Data (U.S. Department of Labor, Office of Foreign Labor Certification, 2016). Author's calculations.

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors clustered by county in parentheses. The dependent variable is log(HHI) in every column. Each column re-estimates $\log(\mathrm{HHI}_{ct}) = \beta\,(\text{Win Rate}_c \times \text{Post}_t) + \delta_c + \gamma_t + \epsilon_{ct}$ on the sample of counties filing at least the stated number of LCA positions. The corresponding log(Shannon) estimates are omitted; they mirror these with the opposite sign at every threshold. A threshold of zero is not attainable, since counties filing no petitions have an undefined win rate, so the sweep begins at one. Column (3) is the main specification and reproduces Table 2 exactly. Thresholds above 50 are not reported: they select so heavily on county size that any movement in the estimate is as easily read as urban heterogeneity as measurement error.

Appendix B: Data

Table A4: Description of Key Variables

Variable Name	Description
`COUNTY`	FIPS code identifier for each U.S. county
`NAME`	Name of each U.S. county
`YEAR`	Year of each observation (2012-2022)
`win_rate`	2016 H-1B lottery win rate by county
`shannon_index`	Shannon index for each county-year
`hhi`	HHI for each county-year
`ESTAB`	Establishment Counts in each county-year
`STATE`	USPS State Abbreviations
`log_shannon`	Shannon index, logged
`log_hhi`	HHI, logged
`Post`	The post-treatment period (2017 and later)
`event_time`	Indicates years from the treatment year (FY2017)
`log_estab`	Establishment Counts, logged

Source: County Business Patterns (U.S. Census Bureau, 2025); H-1B Employer Data Hub (U.S. Citizenship and Immigration Services, 2016); LCA Disclosure Data (U.S. Department of Labor, Office of Foreign Labor Certification, 2016).
Notes: Variables in the analysis dataset, a balanced county-year panel covering 2012–2022. Logged variables are natural logs.

Software Used

RStudio, Stata

R packages

`data.table, readxl, dplyr, lubridate, tidyverse, janitor, tidycensus tigris, stringr, ggplot2, sf, viridis, tidyr, lmtest, sandwich, fixest, modelsummary`

Stata packages

`ftools, reghdfe`

All cleaning scripts and replication files are available at Github Repository